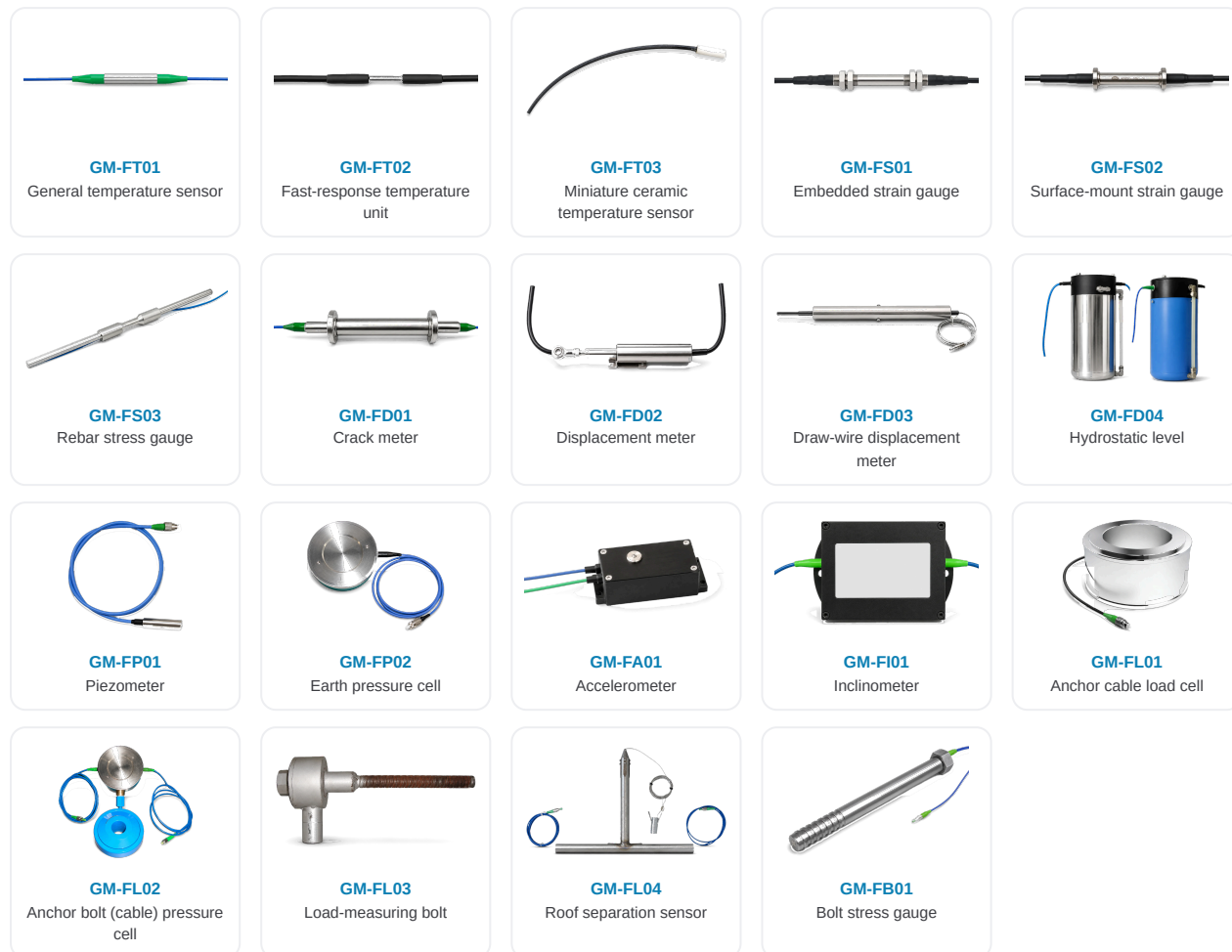


# GM-F Series Fiber Bragg Grating Sensors

8 families · 21 standard models · IP68 across the range · C-band 1528~1568 nm · FC/APC

The GM-F series covers eight physical quantities: temperature, strain, displacement, pressure, vibration, inclination, anchoring and bolts. The sensing end is passive, EMI-immune and intrinsically safe; range, size and packaging are customizable across the range; strain types pair with GM-FT01 for temperature compensation; compatible with third-party C-band interrogators.

## Product family



## Naming & selection guide

Naming rule GM-F-T-01: GM brand · F grating sensor · quantity code · serial number. Quantity codes: T temperature / S strain / D displacement / P pressure / A dynamics / I inclination / L anchoring / B bolt. Three steps: ① pick the family by quantity; ② pick the model by range and mounting; ③ add one GM-FT01 per strain / displacement point for temperature compensation.

## Platform parameters

|                    |                                                  |
|--------------------|--------------------------------------------------|
| Wavelength band    | C-band 1528~1568 nm                              |
| Fiber connector    | FC/APC                                           |
| Ingress protection | IP68 across the range (see table for exceptions) |

## Specification table

| Model                                      | Name                                 | Range                             | Accuracy / resolution       | Key parameters                                                          |
|--------------------------------------------|--------------------------------------|-----------------------------------|-----------------------------|-------------------------------------------------------------------------|
| Temperature T · TEMPERATURE                |                                      |                                   |                             |                                                                         |
| GM-FT01                                    | General temperature sensor           | -40~120 °C (custom)               | ±0.5 °C / 0.1 °C            | Φ9×100 mm · IP68 · clamp / strap / embed                                |
| GM-FT02                                    | Fast-response temperature unit       | -40~120 °C (custom)               | ±0.5 °C / 0.1 °C            | Φ7×100 mm · IP68 · hazardous areas · fire detection                     |
| GM-FT03                                    | Miniature ceramic temperature sensor | -40~120 °C (custom)               | ±0.5 °C / 0.1 °C            | Φ4×45 mm · IP68 · switchgear contacts, tight spots                      |
| Strain & rebar stress S · STRAIN           |                                      |                                   |                             |                                                                         |
| GM-FS01                                    | Embedded strain gauge                | ±1000 / ±1500 με                  | 1% F.S. / 1 με              | 100 mm gauge · embedded · IP68 · pair with FT01 for temp compensation   |
| GM-FS02                                    | Surface-mount strain gauge           | ±500 / 1000 / 1500 με             | 1% F.S. / 1 με              | 100 mm gauge · weld on steel / bond on concrete · IP68                  |
| GM-FS03                                    | Rebar stress gauge                   | 200 MPa tension / 100 compression | 1% F.S. / 1 MPa             | All rebar sizes · 600 mm · butt-weld / strap                            |
| Displacement & settlement D · DISPLACEMENT |                                      |                                   |                             |                                                                         |
| GM-FD01                                    | Crack meter                          | 20 mm (custom)                    | 1 / 0.1 mm                  | Fully sealed embedded · -40~80 °C · IP68                                |
| GM-FD02                                    | Displacement meter                   | 10 / 100 mm                       | 0.1 / 0.5 mm · res. 0.01 mm | Φ40×200 mm · base mount, easy swap & calibration                        |
| GM-FD03                                    | Draw-wire displacement meter         | 100 mm (custom)                   | 1 / 0.1 mm                  | Φ40×350 mm · 20 m wire remote lead standard                             |
| GM-FD04                                    | Hydrostatic level                    | 0~100 mm                          | 1 / 0.1 mm                  | Φ120×250 mm · networked settlement · stainless / molded                 |
| Pressure P · PRESSURE                      |                                      |                                   |                             |                                                                         |
| GM-FP01                                    | Piezometer                           | 0.35 / 1 MPa (custom)             | 0.5% / 0.1% F.S.            | Φ30×120 mm · drop-in / embed · IP68                                     |
| GM-FP02                                    | Earth pressure cell                  | 1 / 2 MPa (custom)                | 1% / 0.1% F.S.              | Custom sizes · embedded · IP68 · pairs with piezometer                  |
| Dynamics A · DYNAMICS                      |                                      |                                   |                             |                                                                         |
| GM-FA01                                    | Accelerometer                        | 2 g · 0.5~50 Hz                   | 2 mg / 0.5 mg · ≈500 pm/g   | 110×35×25 mm · IP68 · standard / enhanced · 2D / 3D combos              |
| Inclination I · INCLINATION                |                                      |                                   |                             |                                                                         |
| GM-FI01                                    | Inclinometer                         | ±5°                               | 0.01° / 0.001°              | 105×80×28 mm · screw mount · IP68 · dual-grating self temp compensation |
| Anchoring & support L · ANCHORING          |                                      |                                   |                             |                                                                         |

|                |                                      |                    |                    |                                                                       |
|----------------|--------------------------------------|--------------------|--------------------|-----------------------------------------------------------------------|
| <b>GM-FL02</b> | Anchor bolt (cable)<br>pressure cell | 0~150 kN           | 5% F.S. · 1 kN     | Sleeved between bolt plate and rock<br>face · IP54                    |
| <b>GM-FL03</b> | Load-measuring bolt                  | 70 kN              | 6% F.S. / 1% F.S.  | Grating inside steel bolt · strain /<br>axial force / bending / shear |
| <b>GM-FL04</b> | Roof separation sensor               | 200 mm, re-armable | 5% F.S.            | 395×306×34 mm · early roof-<br>separation trends                      |
| <b>GM-FL05</b> | Smart bolt                           | 500 MPa tension    | 1% (0.5% optional) | FRP bolt body is the sensor ·<br>embedded grating · IP68              |

**Bolt B · BOLT**

|                |                   |         |                  |                                                           |
|----------------|-------------------|---------|------------------|-----------------------------------------------------------|
| <b>GM-FB01</b> | Bolt stress gauge | 100 MPa | <1% / ≤0.1% F.S. | Custom per bolt size · armored<br>waterproof · -30~+80 °C |
|----------------|-------------------|---------|------------------|-----------------------------------------------------------|

Wind-power products continue the proven supply models (GM-WTB / WTT / WTF series); see the Wind Turbine Monitoring Sensors datasheet.

**Warranty & service**

|                                                      |                                             |                                             |                                           |
|------------------------------------------------------|---------------------------------------------|---------------------------------------------|-------------------------------------------|
| <b>2 yr</b><br>Hardware warranty · spares<br>stocked | <b>24 h</b><br>On-site response, East China | <b>48 h</b><br>On-site response, nationwide | <b>7×24</b><br>Remote diagnostics hotline |
|------------------------------------------------------|---------------------------------------------|---------------------------------------------|-------------------------------------------|

Standard service policy; the supply contract prevails.

**Certifications & standards**

|                                                                                                     |                                                                                                      |                                                                                                         |                                                                                                                  |                                                                                                           |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <br>CNAS testing | <br>CMA metrology | <br>ISO 9001 quality | <br>ISO 14001 environmental | <br>MA mining safety |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

Certificate numbers and scans are available on request.